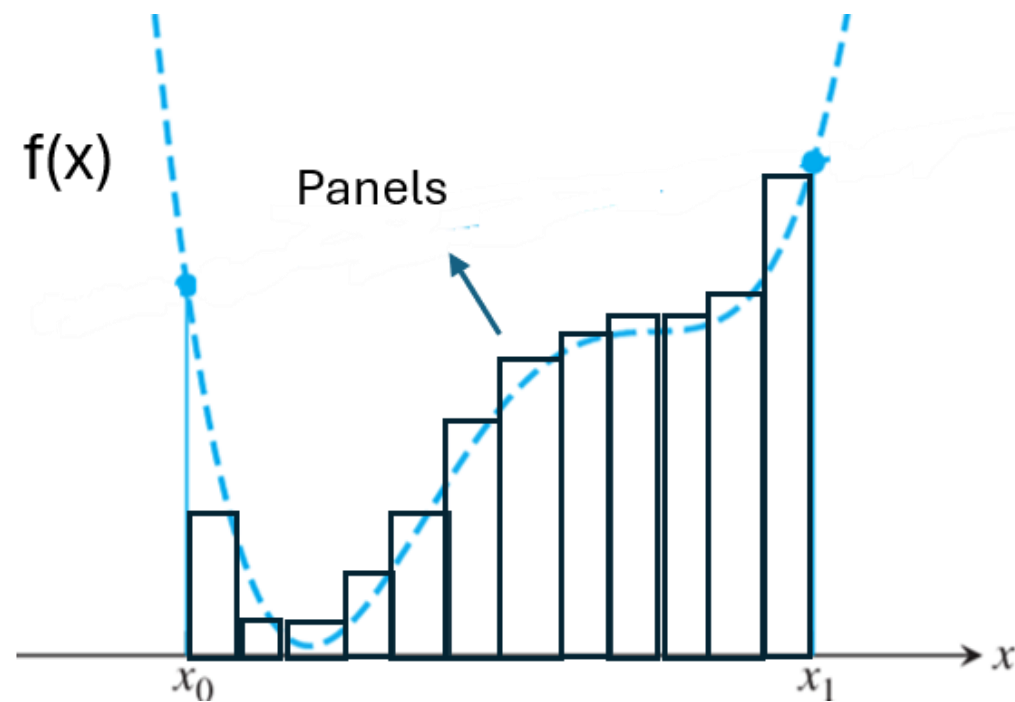
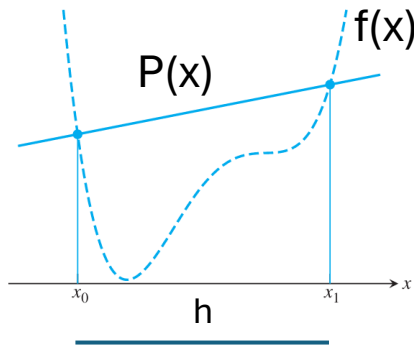


Integration

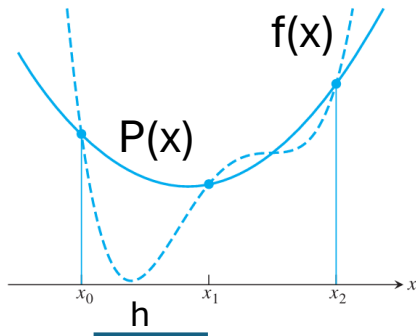


Trapezoid Rule



$$\int_{x_0}^{x_1} f(x) dx = \frac{h}{2}(y_0 + y_1)$$

Simpson's Rule



$$\int_{x_0}^{x_2} f(x) dx = \frac{h}{3}(y_0 + 4y_1 + y_2)$$

(Composite methods for better accuracy.)

Exercise: Numerically evaluate the integral below using trapezoidal and Simpson's method.

$$\int_0^1 x^2 dx \quad \left\{ \frac{x^3}{3} \right|_0^1 = 1/3 \right\}$$

Trapezoid: $h=1$

$$\int_0^1 x^2 dx = \frac{1}{2}(0+1)=1/2, \quad \text{Error: } 1/2-1/3=1/6$$

Error estimate using the formula $\frac{h^3}{12}f''(c)=1/12*2=1/6$

Simpson: $h=1/2$

$$\int_0^1 x^2 dx = \frac{1}{2*3}(0 + 4*1/4 + 1) = 1/3$$

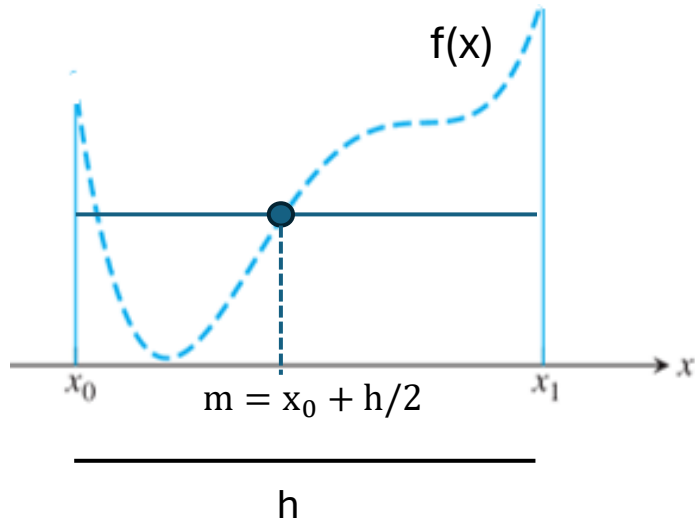
$$\text{Error}=1/3-1/3=0$$

$$\text{Error using } \frac{h^5}{90}f''''(c) = 0$$

(Error is zero because the degree of precision of the Simpson method is **3**)

(work out for $f(x) = xe^x$)

Midpoint Method:



Midpoint rule:

$$\int_{x_0}^{x_1} f(x) dx = hf(m)$$

Use midpoint rule in cases where the integrand has **singularities** at the interval endpoints

$$\int_0^1 \frac{\sin(x)}{x} dx$$

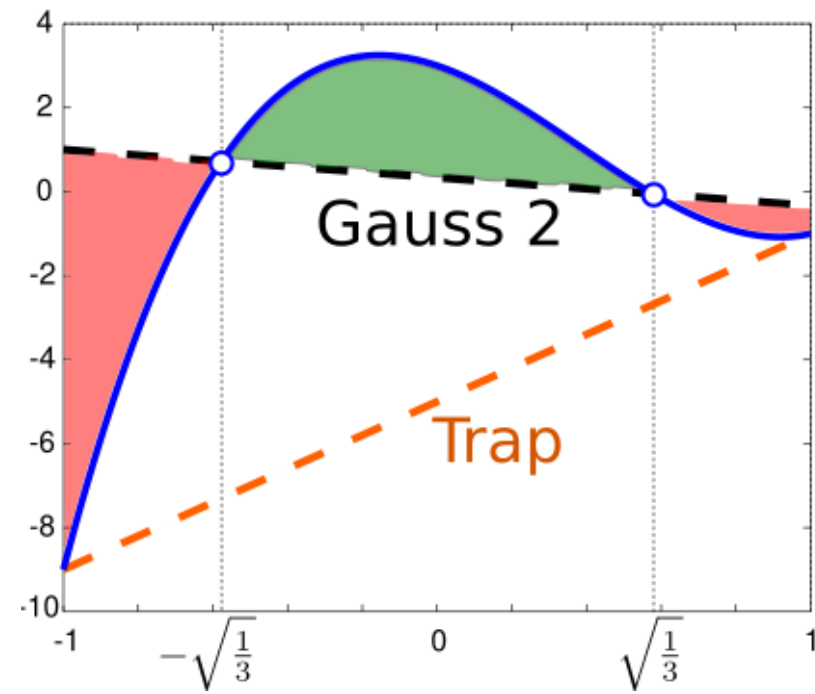
Following the same steps as for the trapezoidal method, show that

Error: $\frac{h^3}{24} f''(c)$, where c is a constant between x_0 and x_1

Composite midpoint rule:

$$\int_a^b f(x) dx = h \sum_{i=1}^n f(m_i) \quad \text{Error: } \frac{(b-a)h^2}{24} f''(c)$$

Gaussian Quadrature Method



Legendre Polynomials

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

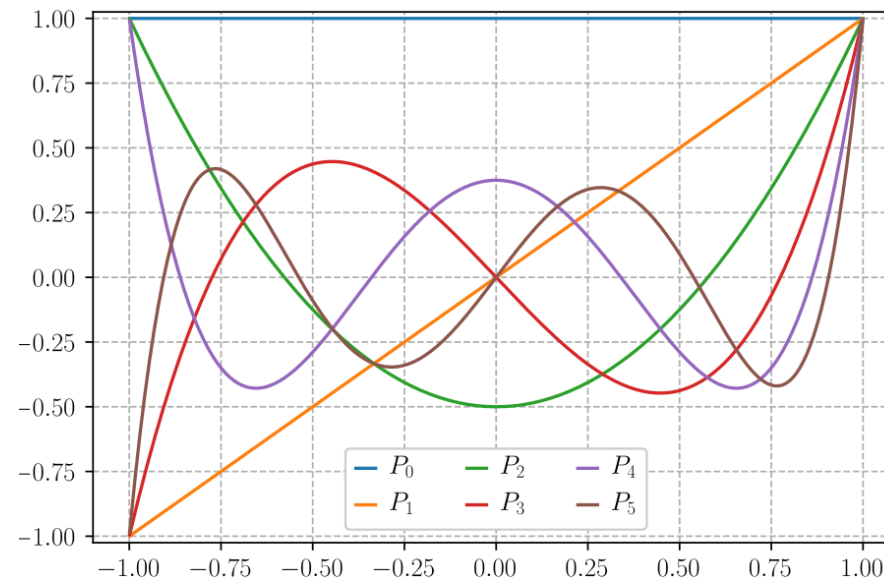
n	$P_n(x)$
0	1
1	x
2	$\frac{1}{2} (3x^2 - 1)$
3	$\frac{1}{2} (5x^3 - 3x)$
4	$\frac{1}{8} (35x^4 - 30x^2 + 3)$

Properties:

$$(1) \quad \int_{-1}^1 P_m(x) P_n(x) dx = 0 \quad \text{if } n \neq m.$$

$$\neq 0 \quad \text{if } n = m$$

$$(2) \quad P_n(x) \text{ has } n \text{ roots in the interval } [-1, 1]$$



Objective: $\int_{-1}^1 f(x) dx$

Choose an interpolating polynomial $Q(x)$

$$Q(x) = \sum_{i=1}^n f(x_i) l_i(x), \quad l_i(x) = \prod_{m=1, m \neq i}^n \frac{(x - x_m)}{(x_i - x_m)}$$

x_i = root of a legendre polynomial

$$\int_{-1}^1 f(x) dx \approx \int_{-1}^1 Q(x) dx = \sum_{i=1}^n c_i f(x_i), \text{ where } c_i = \int_{-1}^1 l_i(x) dx$$

Exercise: $\int_{-1}^1 e^{-\frac{x^2}{2}} dx,$

$n = 2$ Gaussian Quadrature approximation

$$\int_{-1}^1 e^{-\frac{x^2}{2}} dx \approx c_1 f(x_1) + c_2 f(x_2)$$

$$= 1 \cdot f(-\sqrt{1/3}) + 1 \cdot f(\sqrt{1/3}) \approx 1.69296344978123.$$

n	roots x_i	coefficients c_i
2	$-\sqrt{1/3} = -0.57735026918963$ $\sqrt{1/3} = 0.57735026918963$	1 = 1.000000000000000 1 = 1.000000000000000
3	$-\sqrt{3/5} = -0.77459666924148$ 0 = 0.000000000000000 $\sqrt{3/5} = 0.77459666924148$	5/9 = 0.555555555555555 8/9 = 0.888888888888888 5/9 = 0.555555555555555
4	$-\sqrt{\frac{15+2\sqrt{30}}{35}} = -0.86113631159405$ $-\sqrt{\frac{15-2\sqrt{30}}{35}} = -0.33998104358486$ $\sqrt{\frac{15-2\sqrt{30}}{35}} = 0.33998104358486$ $\sqrt{\frac{15+2\sqrt{30}}{35}} = 0.86113631159405$	$\frac{90-5\sqrt{30}}{180} = 0.34785484513745$ $\frac{90+5\sqrt{30}}{180} = 0.65214515486255$ $\frac{90+5\sqrt{30}}{180} = 0.65214515486255$ $\frac{90-5\sqrt{30}}{180} = 0.34785484513745$

(roots are not equispaced)

the $n = 4$ approximation is

$$c_1 f(x_1) + c_2 f(x_2) + c_3 f(x_3) + c_4 f(x_4) \approx 1.71122450459949.$$

The correct answer to 14 digits is 1.71124878378430.

Composite Simpson method with **8000** panels: 1.71129

To approximate integrals in a generic interval $[a, b]$, use the substitution

$$t = (2x - a - b)/(b - a).$$

$$\int_a^b f(x) \, dx = \int_{-1}^1 f\left(\frac{(b-a)t + b + a}{2}\right) \frac{b-a}{2} \, dt.$$

Theorem

If $\{p_0, p_1, \dots, p_n\}$ is an orthogonal set of polynomials on the interval $[a, b]$, where $\deg p_i = i$, then $\{p_0, p_1, \dots, p_n\}$ is a basis for the vector space of degree at most n polynomials on $[a, b]$. 

Proof: We must show that the orthogonal polynomials are linearly independent.

Assume linear dependence, i.e., $\sum_{i=0}^n c_i p_i = 0$, where c_i 's are constants.

$$0 = \int_a^b p_k \sum_{i=0}^n c_i p_i(x) dx = \sum_{i=0}^n c_i \int_a^b p_k p_i dx = c_k \int_a^b p_k^2 dx. \quad \text{Therefore, } c_k = 0.$$

 $\{p_0, p_1, \dots, p_n\}$ is a basis

The Gaussian Quadrature Method, using the degree n Legendre polynomial on $[-1, 1]$, has degree of precision $2n - 1$. ■

Proof. Let $P(x)$ be a polynomial of degree at most $2n - 1$. We must show it is integrated exactly by Gaussian Quadrature.

Using long division of polynomials, we can express

$$P(x) = S(x)p_n(x) + R(x), \quad \text{where } S(x) \text{ and } R(x) \text{ are polynomials of degree less than } n, \text{ and } p_n(x) \text{ is the degree } n \text{ Legendre polynomial}$$

Long division of polynomials: Divide $x^3 - 2x^2 + 0x - 4$ by $(x - 3)$

$$\begin{array}{r} x^2 \\ x-3 \overline{) x^3 - 2x^2 + 0x - 4} \\ \underline{x^3 - 3x^2} \end{array}$$

$$\begin{array}{r} x^2 \\ x-3 \overline{) x^3 - 2x^2 + 0x - 4} \\ \underline{x^3 - 3x^2} \\ +x^2 + 0x \end{array}$$

$$\begin{array}{r} x^2 + x \\ x-3 \overline{) x^3 - 2x^2 + 0x - 4} \\ \underline{x^3 - 3x^2} \\ +x^2 + 0x \\ \underline{+x^2 - 3x} \\ +3x - 4 \end{array}$$

$$\begin{array}{r} x^2 + x + 3 \\ x-3 \overline{) x^3 - 2x^2 + 0x - 4} \\ \underline{x^3 - 3x^2} \\ +x^2 + 0x \\ \underline{+x^2 - 3x} \\ +3x - 4 \\ \underline{+3x - 9} \\ +5 \end{array}$$

$$x^3 - 2x^2 - 4 = (x - 3)(x^2 + x + 3) + 5$$

$$\int_{-1}^1 P(x) dx = \int_{-1}^1 S(x) p_n(x) dx + \int_{-1}^1 R(x) dx = 0 + \int_{-1}^1 R(x) dx,$$

Because **$S(x)$** can be spanned in terms of Legendre polynomials of degree less than **n** , which are orthogonal to **$p_n(x)$** .

A Gaussian quadrature with n points is exact on $R(x)$, because the degree of $R(x)$ is $n-1$.

Therefore the integral of $P(x)$ with degree $2n-1$ is exactly estimated by a Gaussian quadrature with n points.